

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

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Lab Assignment 13: Recursion on Arrays

Experiment No. 13.1 - Recursive array sum

Experiment Title:

Recursive array sum

Aim:

To find the sum of array values from index i to the end using recursion.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

To apply recursion to an array, an extra parameter i holds the current index. Each call handles a[i] and then calls itself with i + 1 (or i - 1) for the rest of the array. The base case is reached when the index goes past the end (or before the start) of the array, or when only one element remains.

Algorithm / Procedure:

1. Base case: if i == a.length return 0.
2. Recursive case: return a[i] + sum(a, i + 1).
3. Call sum(a, 0) from main and print the result.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 13, Task 1
// Question: Write the method int sum(int[] a, int i) that returns
// the sum of the values from index i to the end. Call it on {3, 6,
// 9, 12, 15} with i = 0 and print the result.

public class Main {
    static int sum(int[] a, int i) {
        if (i == a.length) {
            return 0;
        }
        return a[i] + sum(a, i + 1);
    }

    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        System.out.println(sum(a, 0));
    }
}
```

Input:

Array {3, 6, 9, 12, 15}, i = 0.

Output:

```
45
```

Time / Space Complexity: Time O(n); Space O(n).

Result:

The program was compiled and executed successfully. The recursive sum of the array is 45.

Viva-Voce Questions:

- Why is the index passed as a parameter?
- What is the base case here?
- How deep is the recursion for n elements?

Experiment No. 13.2 - Recursive array maximum

Experiment Title:

Recursive array maximum

Aim:

To find the largest value from index i to the end using recursion.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

To apply recursion to an array, an extra parameter i holds the current index. Each call handles a[i] and then calls itself with i + 1 (or i - 1) for the rest of the array. The base case is reached when the index goes past the end (or before the start) of the array, or when only one element remains.

Algorithm / Procedure:

1. Base case: if i is the last index, return a[i].
2. Find rest = max(a, i + 1).
3. Return the larger of a[i] and rest.
4. Call max(a, 0) from main and print the result.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 13, Task 2
// Question: Write the method int max(int[] a, int i) that returns
// the largest value from index i to the end. Call it on {3, 6, 9,
// 12, 15} with i = 0 and print the result.

public class Main {
    static int max(int[] a, int i) {
        if (i == a.length - 1) {
            return a[i];
        }
        int rest = max(a, i + 1);
        if (a[i] > rest) {
            return a[i];
        }
        return rest;
    }

    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        System.out.println(max(a, 0));
    }
}
```

Input:

Array {3, 6, 9, 12, 15}, i = 0.

Output:

```
15
```

Time / Space Complexity: Time O(n); Space O(n).

Result:

The program was compiled and executed successfully. The recursive maximum of the array is 15.

Viva-Voce Questions:

- Why is the base case the last index and not a.length?
- How does the method compare values?
- What is the time complexity?

Experiment No. 13.3 - Print an array in reverse

Experiment Title:

Print an array in reverse

Aim:

To print array values from index i down to 0 using recursion.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

To apply recursion to an array, an extra parameter i holds the current index. Each call handles a[i] and then calls itself with i + 1 (or i - 1) for the rest of the array. The base case is reached when the index goes past the end (or before the start) of the array, or when only one element remains.

Algorithm / Procedure:

1. Base case: if i < 0 return.
2. Print a[i].
3. Call printReverse(a, i - 1).
4. Call it from main with i = a.length - 1.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 13, Task 3
// Question: Write the method void printReverse(int[] a, int i) that
// prints the values from index i down to 0, one per line, using
// recursion. Call it on {3, 6, 9, 12, 15} with i set to the last
// index.

public class Main {
    static void printReverse(int[] a, int i) {
        if (i < 0) {
            return;
        }
        System.out.println(a[i]);
        printReverse(a, i - 1);
    }

    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        printReverse(a, a.length - 1);
    }
}
```

Input:

Array {3, 6, 9, 12, 15}, i = 4.

Output:

```
15
12
9
6
3
```

Time / Space Complexity: Time O(n); Space O(n).

Result:

The program was compiled and executed successfully. The array was printed in reverse order: 15, 12, 9, 6, 3.

Viva-Voce Questions:

- What does a void method return?
- What changes if the print is placed after the recursive call?
- What is the base case?